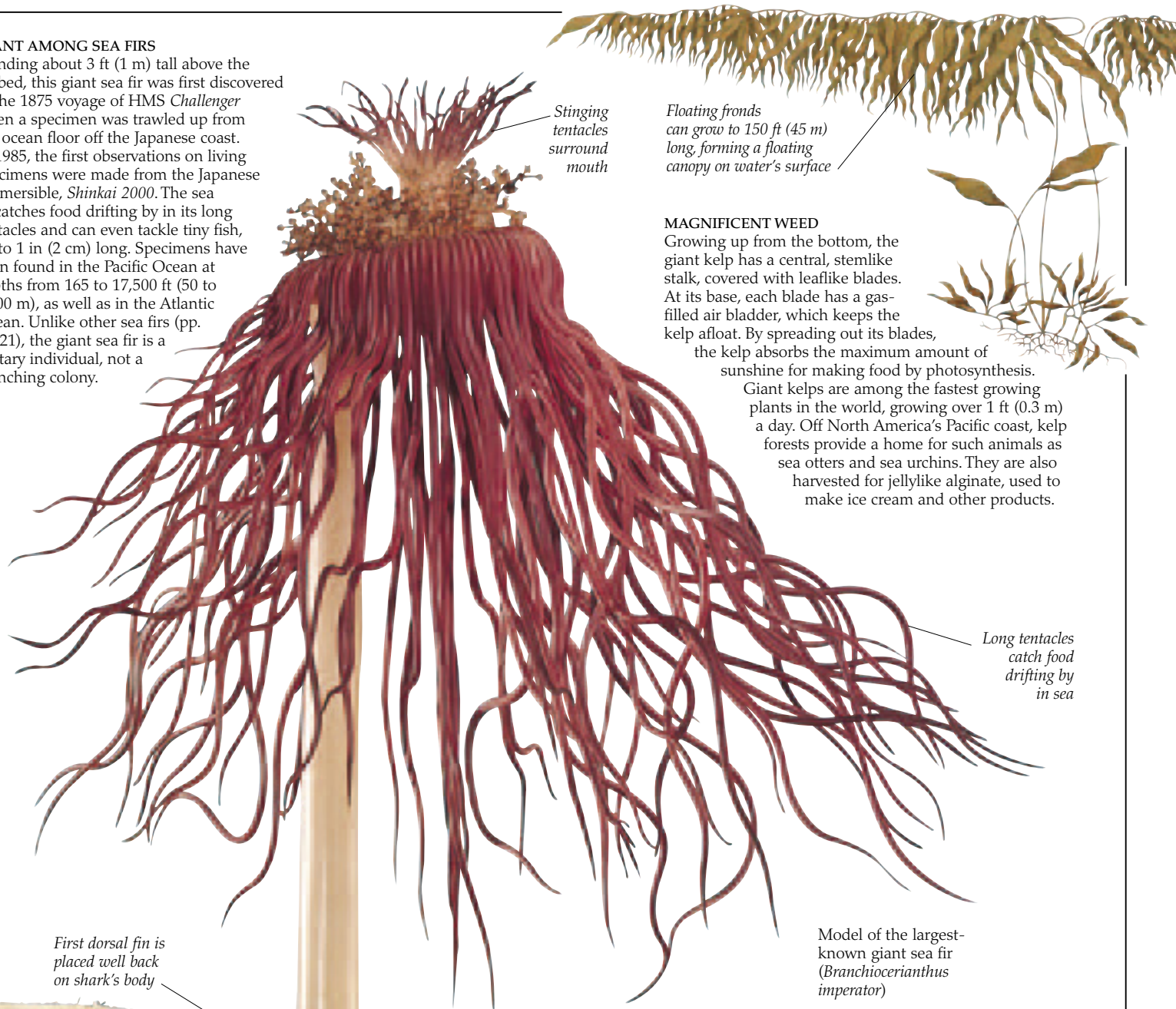


GIANT AMONG SEA FIRS

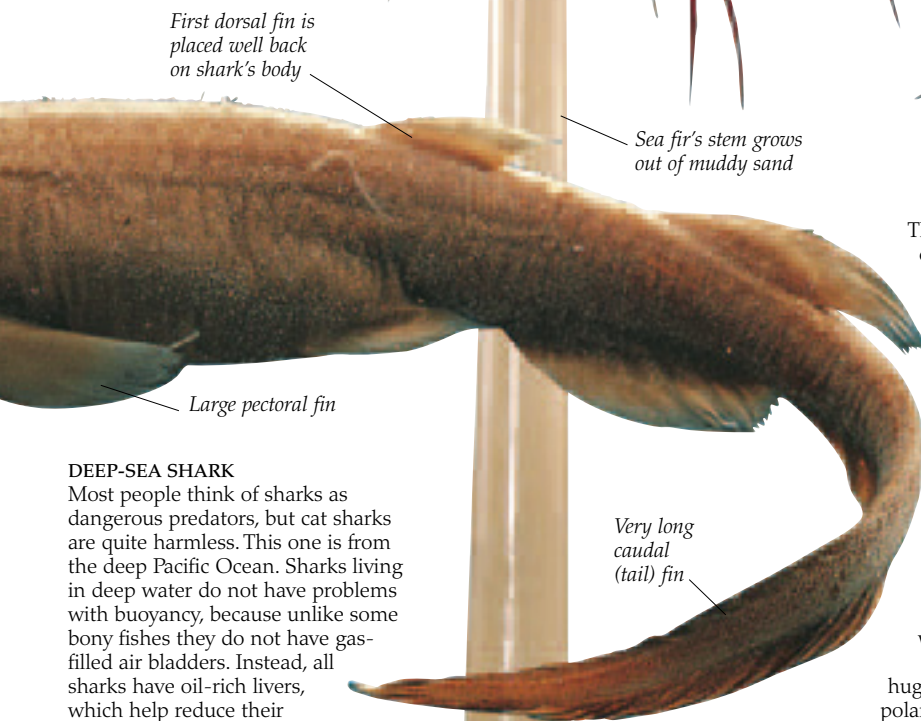
Standing about 3 ft (1 m) tall above the seabed, this giant sea fir was first discovered in the 1875 voyage of HMS *Challenger* when a specimen was trawled up from the ocean floor off the Japanese coast. In 1985, the first observations on living specimens were made from the Japanese submersible, *Shinkai 2000*. The sea fir catches food drifting by in its long tentacles and can even tackle tiny fish, up to 1 in (2 cm) long. Specimens have been found in the Pacific Ocean at depths from 165 to 17,500 ft (50 to 5,300 m), as well as in the Atlantic Ocean. Unlike other sea firs (pp. 20–21), the giant sea fir is a solitary individual, not a branching colony.



MAGNIFICENT WEED

Growing up from the bottom, the giant kelp has a central, stemlike stalk, covered with leaflike blades. At its base, each blade has a gas-filled air bladder, which keeps the kelp afloat. By spreading out its blades, the kelp absorbs the maximum amount of sunshine for making food by photosynthesis.

Giant kelps are among the fastest growing plants in the world, growing over 1 ft (0.3 m) a day. Off North America's Pacific coast, kelp forests provide a home for such animals as sea otters and sea urchins. They are also harvested for jellylike alginate, used to make ice cream and other products.



DEEP-SEA SHARK

Most people think of sharks as dangerous predators, but cat sharks are quite harmless. This one is from the deep Pacific Ocean. Sharks living in deep water do not have problems with buoyancy, because unlike some bony fishes they do not have gas-filled air bladders. Instead, all sharks have oil-rich livers, which help reduce their weight in water.

Model of the largest-known giant sea fir (*Branchiocerianthus imperator*)

ICY OCEAN

There are two main types of sea ice—pack ice that forms on the surface of the open sea (as here in Canada's Hudson Bay) and fast ice that forms between the land and the pack ice. Because of its salt content, seawater freezes at lower temperatures than freshwater. Since cold water sinks and is replaced by warmer water, it takes a long time before the water is cold enough for ice to form. Icebergs are huge chunks broken off the polar ice sheets and glaciers, formed from freshwater on land.

